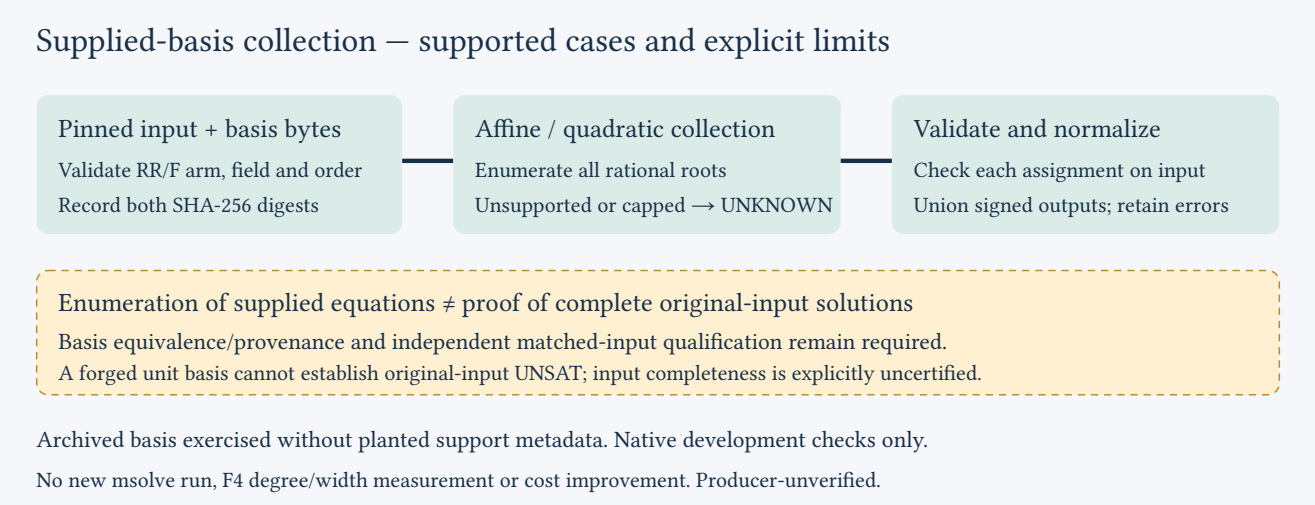
Supplied-basis collection for RR/F qualification

12 November 2026 · Native development checks · Producer-unverified

The native collector enumerates rational assignments of supplied equations that reduce through affine pivots and univariate quadratics, validates each assignment through the input/extractor, and unions signed decompositions. Unsupported, free-variable and capped residuals remain unknown. Twelve native tests pass. Claim PCC-20261112-72ed05 is unverified.



Supported scope and provenance boundary

The collector validates the RR/F arm, exact prime field, variable order and input digest before enumeration, including empty results. It records the basis digest. Constants can prove the supplied equations inconsistent; quadratic branches enumerate distinct rational roots and affine pivots reconstruct the other coordinates. Resource caps and any unresolved branch produce unknown, never a truncated complete answer. Normalization errors remain visible.

Completeness is relative to the supplied equations. The collector does not certify that this basis generates the original input ideal. Its output always marks original-input completeness uncertified. A forged unit basis therefore cannot establish original-input UNSAT. Exit zero alone does not admit a timing comparison or prove input completeness.

Development checks

The archived p1151 m3t4 F output contains affine relations and one quadratic. The corresponding input and public metadata are reused, after removing the planted coordinate map, support x/y/w values, separator values and seed data. Both rational assignments are recovered and normalized without rerunning msolve. Synthetic fixtures compare affine/quadratic assignments against exhaustive small-field evaluation and distinguish contradiction, unsupported coupled or higher-degree cases, resource caps, and partially supported branches. Negative header, input-hash and empty-arm cases are checked. Read-only review found no remaining concrete bug after the metadata-purge and arm-validation fixes.

Remaining original objective

Before matched F4 cost measurements, bind each basis to its exact source-built solver execution and qualify complete input sets against an independent reference. General unsupported forms and excluded charts remain open. This addition does not measure or predict retention of RR's lower solving degree, smaller matrices or lower complete cost against F.

Checkpoint upload transport and live shared budget/pool readback remain absent. Existing GitHub experiment and fixed-diagnosis workflows were inspected; neither provides a suitable read-only checkpoint/budget preflight. No workflow, build or paid experiment was dispatched. A nonsecret existing runner connection was requested while native work continued. The frozen-ledger merge dependency in PR 3835 is also unchanged.

Artifacts and sources

COLLECTION-20261112.md specifies statuses and commands; the final native log is checks/collection-tests-final-20261112.txt. Source: p1tool/src/collection.rs. The existing P1 input/basis archives and parser define the concrete format. Official msolve documentation distinguishes reduced grevlex basis output from rational assignment collection. KB remains unavailable. Historical reports/scoreboards are unchanged. PR 3835.